

PSET 2: Limits and Derivatives

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Limits: left, right, and at a point

Given that¹

$$\lim_{x \rightarrow a} f(x) = -3, \quad \lim_{x \rightarrow a} g(x) = 0, \quad \lim_{x \rightarrow a} h(x) = 8$$

find the limits that exist. If a limit does not exist, explain why.

a. $\lim_{x \rightarrow a} [f(x) + h(x)]$

Ans.:

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) + h(x)] &= \lim_{x \rightarrow a} [f(x)] + \lim_{x \rightarrow a} [h(x)] \\ &= -3 + 8 \\ &= 5 \end{aligned}$$

Limit exists.

b. $\lim_{x \rightarrow a} \frac{g(x)}{f(x)}$

Ans.:

$$\begin{aligned} \lim_{x \rightarrow a} \frac{g(x)}{f(x)} &= \frac{\lim_{x \rightarrow a} g(x)}{\lim_{x \rightarrow a} f(x)} \\ &= \frac{0}{-3} \\ &= 0 \end{aligned}$$

Limit exists.

c. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$

¹Grimmer 2012 HW 2.4

Ans.:

$$\begin{aligned}\lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \\ &= \frac{-3}{0}\end{aligned}$$

Limit does not exist since the limit is indeterminable (division by 0).

d. $\lim_{x \rightarrow a} \frac{2f(x)}{h(x) - f(x)}$

Ans.:

$$\begin{aligned}\lim_{x \rightarrow a} \frac{2f(x)}{h(x) - f(x)} &= \frac{\lim_{x \rightarrow a} 2f(x)}{\lim_{x \rightarrow a} h(x) - f(x)} \\ &= \frac{2 \times \lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} h(x) - \lim_{x \rightarrow a} f(x)} \\ &= \frac{-6}{8 - (-3)} \\ &= -\frac{6}{11}\end{aligned}$$

Limit exists.

Problem 2: Find even more limits

Find the following limits:²

a. $\lim_{x \rightarrow -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4}$

Ans.:

$$\begin{aligned}\lim_{x \rightarrow -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4} &= \frac{(-4)^2 + 5(-4) + 4}{(-4)^2 + 3(-4) - 4} \\ &= \frac{16 - 20 + 4}{16 - 12 - 4} \\ &= \frac{0}{0}\end{aligned}$$

Indeterminable form, so need to factor.

$$\begin{aligned}\lim_{x \rightarrow -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4} &= \lim_{x \rightarrow -4} \frac{x^2 + x + 4x + 4}{x^2 - x + 4x - 4} \\ &= \lim_{x \rightarrow -4} \frac{x(x+1) + 4(x+1)}{x(x-1) + 4(x-1)} \\ &= \lim_{x \rightarrow -4} \frac{(x+1)(x+4)}{(x-1)(x+4)} \\ &= \lim_{x \rightarrow -4} \frac{(x+1)}{(x-1)}\end{aligned}$$

$$\therefore \lim_{x \rightarrow -4} \frac{x^2 + 5x + 4}{x^2 + 3x - 4} = \frac{-3}{-5} = \frac{3}{5}$$

²Grimmer 2012 HW 2.4(b)

b. $\lim_{x \rightarrow 4^-} \sqrt{16 - x^2}$

Ans.:

$$\begin{aligned}\lim_{x \rightarrow 4^-} \sqrt{16 - x^2} &= \sqrt{16 - (4)^2} \\ &= \sqrt{16 - 16}\end{aligned}$$

$$\boxed{\therefore \lim_{x \rightarrow 4^-} \sqrt{16 - x^2} = 0}$$

Problem 3: Check for discontinuities

Which of the following functions are continuous? If a function is not continuous, where are the discontinuities?³

a. $f(x) = \frac{9x^3 - x}{(x - 1)(x + 1)}$

Ans.: Discontinuous at $x = 1, -1$ since denominator becomes 0 (indeterminable limit). Can not be factored any further either.

b. $f(x) = e^{-x^2}$

Ans.: Continuous for all real values of x.

c. $f(x) = \begin{cases} x^3 + 1, & x > 0 \\ -x^2, & x \leq 0 \end{cases}$

Ans.:

$$\begin{aligned}\lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} (-x^2) \\ &= -0^2 \\ &= 0\end{aligned}$$

$$\begin{aligned}\lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} [x^3 + 1] \\ &= 0^3 + 1 \\ &= 1\end{aligned}$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) \neq \lim_{x \rightarrow 0^+} f(x)$$

So, $f(x)$ is not continuous, with a discontinuity at $x = 0$.

Problem 4: Find finite limits

Find the following finite limits:⁴

³Gill 1.9

⁴Gill 5.1

a. $\lim_{x \rightarrow 1} \left[\frac{x^4 - 1}{x - 1} \right]$

Ans.:

$$\begin{aligned} \lim_{x \rightarrow 1} \left[\frac{x^4 - 1}{x - 1} \right] &= \lim_{x \rightarrow 1} \left[\frac{(x^2 + 1)(x^2 - 1)}{x - 1} \right] \\ &= \lim_{x \rightarrow 1} \left[\frac{(x^2 + 1)(x + 1)(x - 1)}{x - 1} \right] \\ &= \lim_{x \rightarrow 1} [(x^2 + 1)(x + 1)] \\ &= (1^2 + 1)(1 + 1) \end{aligned}$$

$$\boxed{\therefore \lim_{x \rightarrow 1} \left[\frac{x^4 - 1}{x - 1} \right] = 2}$$

Problem 5: Find infinite limits

Find the following infinite limits:⁵

Hint: use **L'Hôpital's rule** to switch from $\lim_{x \rightarrow \infty} \left(\frac{f(x)}{g(x)} \right)$ to $\lim_{x \rightarrow \infty} \left(\frac{f'(x)}{g'(x)} \right)$.

a. $\lim_{x \rightarrow \infty} \left[\frac{2^x - 3}{2^x + 1} \right]$

Ans.:

$$\begin{aligned} \lim_{x \rightarrow \infty} \left[\frac{2^x - 3}{2^x + 1} \right] &= \lim_{x \rightarrow \infty} \left[\frac{(2^x - 3)'}{(2^x + 1)'} \right] \\ &= \lim_{x \rightarrow \infty} \left[\frac{2^x \log(2) - 0}{2^x \log(2) + 0} \right] \\ &= \lim_{x \rightarrow \infty} \left[\frac{2^x \log(2)}{2^x \log(2)} \right] \\ &= \lim_{x \rightarrow \infty} 1 \end{aligned}$$

$$\boxed{\therefore \lim_{x \rightarrow \infty} \left[\frac{2^x - 3}{2^x + 1} \right] = 1}$$

b. $\lim_{x \rightarrow \infty} \left[\frac{3^x}{x^3} \right]$

⁵Gill 5.3 and 5.8

Ans.:

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \left[\frac{3^x}{x^3} \right] &= \lim_{x \rightarrow \infty} \left[\frac{(3^x)'}{(x^3)'} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{3^x \log(3)}{3x^2} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{(3^x \log(3))'}{(3x^2)'} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{3^x \log(3) \times \log(3)}{3 \times 2x} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{(3^x \log(3) \log(3))'}{(6x)'} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{3^x \log(3) \log(3) \log(3)}{6} \right] \\
 &= \frac{3^\infty \log(3) \log(3) \log(3)}{6}
 \end{aligned}$$

$$\boxed{\therefore \lim_{x \rightarrow \infty} \left[\frac{3^x}{x^3} \right] = \infty}$$

Problem 6: Assess continuity and differentiability

For each of the following functions, state whether it is continuous and/or differentiable at the point where its two formulas meet.⁶

a. $f(x) = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$

Ans.:

- i. Checking for Continuity (left-hand limit = right-hand limit)

$$\begin{aligned}
 \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} x^2 = 0 \\
 \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} -x^2 = 0 \\
 \therefore \lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^+} f(x)
 \end{aligned}$$

So, $f(x)$ is continuous.

- ii. Checking for Differentiability (left-hand derivative = right-hand derivative, or no abrupt changes)

$$\begin{aligned}
 \lim_{x \rightarrow 0^-} f'(x) &= \lim_{x \rightarrow 0^-} (2x) = 2 \times 0 = 0 \\
 \lim_{x \rightarrow 0^+} f'(x) &= \lim_{x \rightarrow 0^+} (-2x) = -2 \times 0 = 0 \\
 \therefore \lim_{x \rightarrow 0^-} f'(x) &= \lim_{x \rightarrow 0^+} f'(x)
 \end{aligned}$$

So, $f(x)$ is differentiable.

⁶Simon and Blume 2.16

$$\text{b. } f(x) = \begin{cases} x^3, & x \leq 1 \\ x, & x > 1 \end{cases}$$

Ans.:

- i. Checking for Continuity (left-hand limit = right-hand limit)

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (x^3) = 1 \\ \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (x) = 1 \\ \therefore \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^+} f(x) \end{aligned}$$

So, $f(x)$ is continuous.

- ii. Checking for Differentiability (left-hand derivative = right-hand derivative, or no abrupt changes)

$$\begin{aligned} \lim_{x \rightarrow 1^-} f'(x) &= \lim_{x \rightarrow 1^-} (3x^2) = 3 \\ \lim_{x \rightarrow 1^+} f'(x) &= \lim_{x \rightarrow 1^+} (1) = 1 \\ \therefore \lim_{x \rightarrow 1^-} f'(x) &\neq \lim_{x \rightarrow 1^+} f'(x) \end{aligned}$$

So, $f(x)$ is not differentiable.

Problem 7: Calculate derivatives

Recall the differentiation rules:

$$\begin{aligned} [kf(x)]' &= kf'(x) && \text{constant multiple} \\ [f(x) \pm g(x)]' &= f'(x) \pm g'(x) && \text{sum} \\ [f(x)g(x)]' &= f'(x)g(x) + f(x)g'(x) && \text{product} \\ \left[\frac{f(x)}{g(x)} \right]' &= \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}, \quad g(x) \neq 0 && \text{quotient} \\ [x^k]' &= kx^{k-1} && \text{power} \end{aligned}$$

Differentiate the following functions:⁷

a. $f(x) = 4x^3 + 2x^2 + 5x + 11$

Ans.:

$$f'(x) = 4 \times 3x^{3-1} + 2 \times 2x^{2-1} + 5 \times 1x^{1-1} + 0$$

$$\therefore f'(x) = 12x^2 + 4x + 5$$

b. $y = \sqrt{30}$

Ans.:

$$\therefore f'(x) = 0$$

⁷Grimmer HW2.3

c. $h(t) = \log(9t + 1)$

Ans.:

$$\begin{aligned} f'(x) &= \frac{1}{9t+1} \times (9t+1)' \\ &= \frac{9+0}{9t+1} \end{aligned}$$

$$\boxed{\therefore f'(x) = \frac{9}{9t+1}}$$

d. $f(x) = \log(x^2 e^x)$

Ans.:

$$\begin{aligned} f'(x) &= \frac{1}{x^2 e^x} \times (x^2 e^x)' \\ &= \frac{1}{x^2 e^x} \times ((x^2)' e^x + x^2 (e^x)') \\ &= \frac{1}{x^2 e^x} \times [(x^2)' e^x + x^2 (e^x)'] \\ &= \frac{1}{x^2 e^x} \times [2x e^x + x^2 e^x] \\ &= \frac{x e^x [2 + x]}{x^2 e^x} \end{aligned}$$

$$\boxed{\therefore f'(x) = \frac{2+x}{x}}$$

e. $h(y) = \left(\frac{1}{y^2} - \frac{3}{y^4} \right) (y + 5y^3)$

Ans.:

$$\begin{aligned} f(x) &= (y^{-2} - 3y^{-4})(y + 5y^3) \\ &= (y^{-2})(y + 5y^3) - (3y^{-4})(y + 5y^3) \\ &= y^{-1} + 5y - 3y^{-3} - 15y^{-1} \\ &= 5y - 3y^{-3} - 14y^{-1} \\ f'(x) &= 5 \times 1y^{1-1} - 3 \times (-3)y^{-3-1} - 14 \times (-1)y^{-1-1} \\ &= 5 + 9y^{-4} + 14y^{-2} \end{aligned}$$

$$\boxed{\therefore f'(x) = 5 + \frac{9}{y^4} + \frac{14}{y^2}}$$

f. $g(t) = \frac{3t-1}{2t+1}$

Ans.:

$$\begin{aligned}g'(x) &= \frac{(3t-1)'(2t+1) - (3t-1)(2t+1)'}{(2t+1)^2} \\&= \frac{(3)(2t+1) - (3t-1)(2)}{(2t+1)^2} \\&= \frac{(6t+3) - (6t-2)}{(2t+1)^2} \\&= \frac{6t+3-6t+2}{(2t+1)^2} \\&= \frac{5}{(2t+1)^2}\end{aligned}$$

$\therefore g'(x) = \frac{5}{(2t+1)^2}$

Problem 8: Use the product and quotient rules

Differentiate the following function twice—once using the product rule, and once using the quotient rule:⁸

$$f(x) = \frac{x^2 - 2x}{x^4 + 6}$$

Ans.:

i. Using Product Rule

$$\begin{aligned}f'(x) &= (x^2 - 2x)'(x^4 + 6)^{-1} + (x^2 - 2x)((x^4 + 6)^{-1})' \\&= \frac{(2x - 2)}{x^4 + 6} + (x^2 - 2x) \cdot \frac{-1}{(x^4 + 6)^2} \cdot (4x^3 + 0) \\&= \frac{2(x - 1)}{x^4 + 6} - \frac{(x^2 - 2x)(4x^3)}{(x^4 + 6)^2} \\&\therefore f'(x) = \frac{2(x - 1)(x^4 + 6) - 4x^4(x - 2)}{(x^4 + 6)^2}\end{aligned}$$

ii. Using Quotient Rule

$$\begin{aligned}f'(x) &= \frac{(x^2 - 2x)'(x^4 + 6) - (x^2 - 2x)(x^4 + 6)'}{(x^4 + 6)^2} \\&= \frac{(2x - 2)(x^4 + 6) - (x^2 - 2x)(4x^3)}{(x^4 + 6)^2} \\&\therefore f'(x) = \frac{2(x - 1)(x^4 + 6) - 4x^4(x - 2)}{(x^4 + 6)^2}\end{aligned}$$

Problem 9: Logarithms and exponential functions

Compute the derivative of each of the following functions:⁹

⁸Grimmer HW2.4

⁹Simon and Blume 5.8

a. $f(x) = \frac{x}{e^x}$

Ans.:

$$\begin{aligned} f(x) &= xe^{-x} \\ f'(x) &= (x)'(e^{-x}) + (x)(e^{-x})' \\ &= (1)(e^{-x}) + (x)(e^{-x} \times (-x)') \\ &= (e^{-x}) + (x)(-e^{-x}) \end{aligned}$$

$$\boxed{f'(x) = e^{-x} - xe^{-x}}$$

b. $h(x) = \frac{x}{\log(x)}$

Ans.:

$$\begin{aligned} h'(x) &= \frac{(x)'(\log(x)) - (x)(\log(x))'}{(\log(x))^2} \\ &= \frac{(1)(\log(x)) - (x)(\frac{1}{x})}{\log^2(x)} \end{aligned}$$

$$\boxed{\therefore h'(x) = \frac{\log(x) - 1}{\log^2(x)}}$$

Problem 10: Composite functions

For each of the following pairs of functions $g(x)$ and $h(z)$, write out the composite functions $g(h[z])$ and $h(g[x])$. In each case, describe the domain of the composite function.¹⁰

a. $g(x) = x^2 + 4, \quad h(z) = 5z - 1$

Ans.:

$$\begin{aligned} g(h[z]) &= (5z - 1)^2 + 4 \\ &= 25z^2 - 10z + 5 \end{aligned} \quad \rightarrow \quad \text{Domain is } \mathbb{R} \text{ or } (-\infty, +\infty)$$

$$\begin{aligned} h(g[x]) &= 5(x^2 + 4) - 1 \\ &= 5x^2 + 19 \end{aligned} \quad \rightarrow \quad \text{Domain is } \mathbb{R} \text{ or } (-\infty, +\infty)$$

b. $g(x) = x^3, \quad h(z) = (z - 1)(z + 1)$

Ans.:

$$\begin{aligned} g(h[z]) &= (z^2 - 1)^3 \\ &= z^6 - 3z^4 + 3z^2 - 1 \end{aligned} \quad \rightarrow \quad \text{Domain is } \mathbb{R} \text{ or } (-\infty, +\infty)$$

$$\begin{aligned} h(g[x]) &= (x^3 - 1)(x^3 + 1) \\ &= x^6 - 1 \end{aligned} \quad \rightarrow \quad \text{Domain is } \mathbb{R} \text{ or } (-\infty, +\infty)$$

¹⁰Simon and Blume 4.1

Problem 11: Chain rule

Use the chain rule to compute the derivative of the composite functions in the previous problem from the derivatives of the two component functions. Then compute each derivative directly, using your expression for the composite function. Simplify and compare your answers.¹¹

a. $g(x) = x^2 + 4, \quad h(z) = 5z - 1$

Ans.:

i. Using Chain Rule:

$$\begin{aligned} g'(h[z]) &= ((5z - 1)^2 + 4)' \\ &= 2(5z - 1)^{2-1} \times (5z - 1)' + (4)' \\ &= (10z - 2)(5) + 0 \\ &= 50z - 10 \end{aligned}$$

$$\begin{aligned} h'(g[x]) &= (5(x^2 + 4) - 1)' \\ &= 5(2x + 0) - 0 \\ &= 10x \end{aligned}$$

ii. Using Direct Computation:

$$\begin{aligned} g'(h[z]) &= 25x^2 - 10x + 5 \\ &= 25 \times 2x - 10 \times 1 + 0 \\ &= 50x - 10 \end{aligned}$$

$$\begin{aligned} h'(g[x]) &= 5x^2 + 19 \\ &= 5 \times 2x + 0 \\ &= 10x \end{aligned}$$

b. $g(x) = x^3, \quad h(z) = (z - 1)(z + 1)$

Ans.:

¹¹Simon and Blume 4.3

i. Using Chain Rule:

$$\begin{aligned}
 g'(h[z]) &= (z^2 - 1)^3 \\
 &= 3(z^2 - 1)^2 \times (z^2 - 1)' \\
 &= 3(z^2 - 1)^2 \times (2z - 0) \\
 &= 6z(z^4 - 2z^2 + 1) \\
 &= 6z^5 - 12z^3 + 6z
 \end{aligned}$$

$$\begin{aligned}
 h'(g[x]) &= [(x^3 - 1)(x^3 + 1)] \\
 &= (x^3 - 1)'(x^3 + 1) + (x^3 - 1)(x^3 + 1)' \\
 &= (3x^2 - 0)(x^3 + 1) + (x^3 - 1)(3x^2 + 0) \\
 &= 3x^2(x^3 + 1) + 3x^2(x^3 - 1) \\
 &= 3x^2(2x^3) \\
 &= 6x^5
 \end{aligned}$$

ii. Using Direct Computation:

$$\begin{aligned}
 g'(h[z]) &= [(z^6 - 3z^4 + 3z^2 - 1)]' \\
 &= 6z^5 - 3 \times 4z^3 + 3 \times 2z - 0 \\
 &= 6z^5 - 12z^3 + 6z
 \end{aligned}$$

$$\begin{aligned}
 h'(g[x]) &= [x^6 - 1] \\
 &= 6x^5 - 0 \\
 &= 6x^5
 \end{aligned}$$

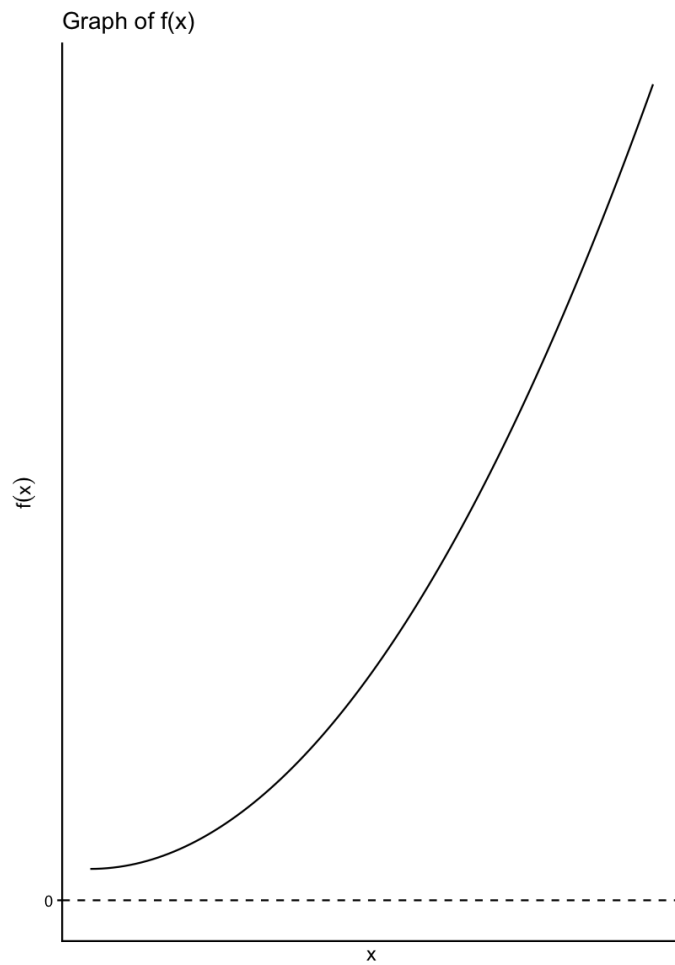
Problem 12: Identify a possible derivative

A friend shows you the graph of a function $f(x)$ below.¹² Which of the following could be a graph of $f'(x)$? For each graph, explain why it might or might not be the derivative of $f(x)$.

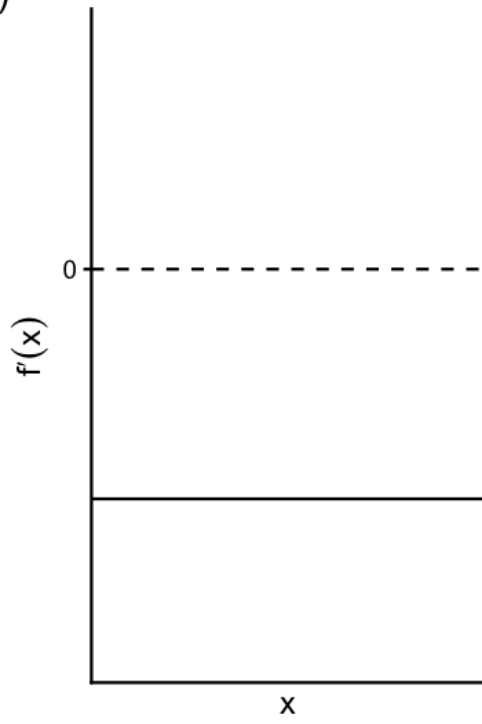
Ans.:

From the graph of $f(x)$, we can see that $f(x)$ is an increasing function that increases increasingly. So, it should have a positive derivative that is sloped upwards. Graph A is negative with zero slope, so it represents a linearly decreasing function - this does not capture $f(x)$. Graph B is positive but without slope, so it represents a linearly increasing function - this does not capture $f(x)$. Graph C is positive and increasing in slope, so it represents an increasingly increasing function - this perfectly describes $f(x)$. Graph D shows an increasing slope of a negative and positive derivative, so it represents a function that is decreasing at first, attains zero slope, then starts increasing increasingly - this does not represent $f(x)$. **Thus, it is Graph C that could be a graph for $f'(x)$.**

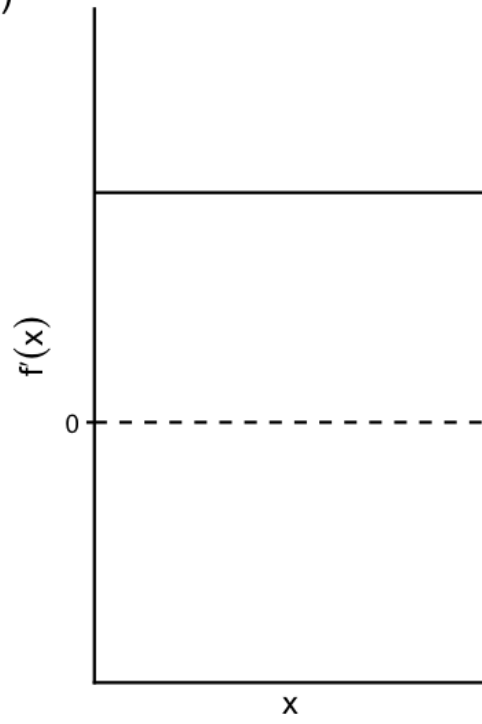
¹²Grimmer HW3.6



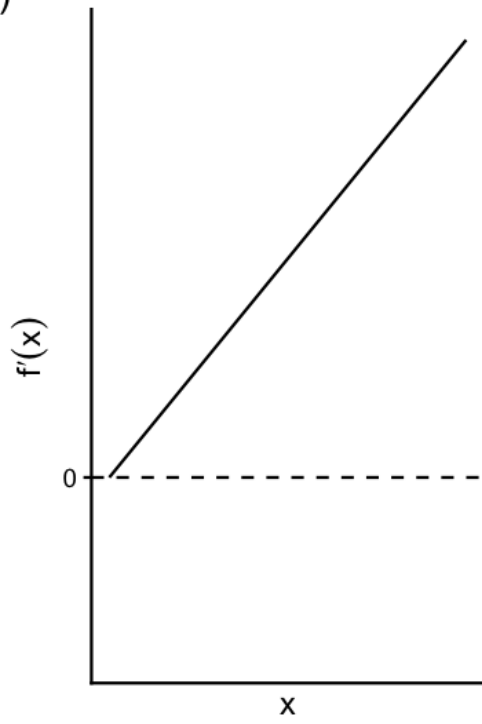
A)



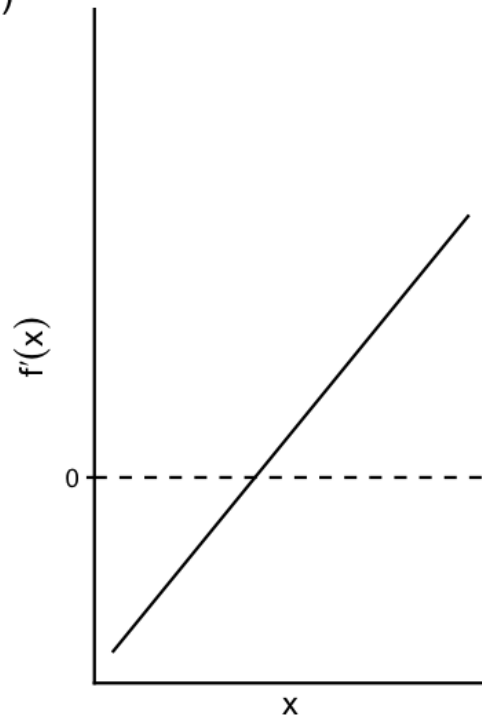
B)



C)



D)



AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.